

Department  
of  
Electronics and Communication Engineering

Minor Project Report  
on

# Design and Implementation of a 16 MHz–1024 MHz Phase-Locked Loop

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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION  
ENGINEERING**

## **CERTIFICATE**

This is to certify that project entitled “ **Design and Implementation of a 16 MHz-1024 MHz Phase-Locked Loop** ” is a bonafide work carried out by the student team of “**Pratiksha N (01FE23BEC310), Sneha B Navalagund (01FE23BEC321), Shivani I Metri (01FE23BEC337), Arpita J Chougule (01FE23BEC357)**”. The project report has been approved as it satisfies the requirements with respect to the minor project work prescribed by the university curriculum for BE (VI Semester) in the School of Electronics and Communication Engineering of KLE Technological University for the academic year 2025-2026.

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# ABSTRACT

A Phase Locked Loop (PLL) is a fundamental circuit widely used in modern VLSI systems for clock generation, frequency synthesis, and synchronization in communication and digital applications. This project presents the design and simulation of a Phase Locked Loop (PLL) using UMC 180 nm CMOS technology in the Cadence Virtuoso design environment. The proposed PLL consists of a Phase Frequency Detector (PFD), Charge Pump, Passive Loop Filter, Voltage Controlled Oscillator (VCO), and a Divide-by-64 Frequency Divider. Each functional block was designed at the transistor level, simulated individually, and then integrated to form the complete PLL. The designed PLL operates with a 1.8 V supply voltage and uses a 16 MHz reference clock to generate an output frequency of approximately 1.024 GHz. The Voltage Controlled Oscillator achieved a gain ( $K_{vco}$ ) of 514 MHz/V, and the Charge Pump was designed with a current of 145 A to ensure stable loop operation. The complete PLL was successfully simulated, and the results confirmed frequency locking with an output frequency of approximately 1.02465 GHz. The proposed design demonstrates reliable frequency multiplication and stable closed-loop operation, making it suitable for clock generation and frequency synthesis applications in CMOS-based VLSI systems. The project also provides practical experience in the transistor-level design and simulation of analog and mixed-signal circuits using the Cadence Virtuoso platform.

Keywords: Phase Locked Loop (PLL), Phase Frequency Detector, Charge Pump, Loop Filter, Voltage Controlled Oscillator, Frequency Divider, CMOS, Cadence Virtuoso.

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# Chapter 1

## Introduction

The rapid advancement of Very Large Scale Integration (VLSI) technology has significantly increased the demand for high-speed, low-power, and reliable clock generation circuits. Modern electronic systems such as communication devices, microprocessors, data converters, and digital signal processing units require precise timing and synchronization to ensure accurate operation. A Phase Locked Loop (PLL) is one of the most important mixed-signal circuits used to generate stable clock signals, perform frequency multiplication, and maintain phase synchronization between different system components.

A PLL is a closed-loop feedback system that continuously compares the phase and frequency of a reference signal with a feedback signal derived from its output. Based on the detected phase difference, the loop automatically adjusts the frequency of the Voltage Controlled Oscillator (VCO) until both signals are synchronized. This closed-loop operation enables the PLL to provide stable output frequencies with good timing accuracy, making it an essential component in modern integrated circuits.

In this project, a Phase Locked Loop is designed and simulated using UMC 180 nm CMOS technology in the Cadence Virtuoso design environment. The proposed design consists of a Phase Frequency Detector (PFD), Charge Pump, Passive Loop Filter, Voltage Controlled Oscillator (VCO), and a Divide-by-64 Frequency Divider. The PLL is designed to generate an output frequency of approximately 1.024 GHz from a 16 MHz reference clock. Each building block is implemented at the transistor level, individually verified through simulation, and integrated to evaluate the overall performance of the complete PLL.

### 1.1 Motivation

The increasing demand for high-speed communication systems and advanced digital integrated circuits has created a need for efficient clock generation and synchronization techniques. Modern applications such as wireless communication, microprocessors, data

converters, and high-speed interfaces require stable and accurate clock signals to ensure reliable system performance. As operating frequencies continue to increase, the design of robust and efficient Phase Locked Loops (PLLs) has become an essential aspect of VLSI system design.

A PLL integrates both analog and digital circuit blocks, making it an excellent platform for understanding mixed-signal circuit design. Designing a PLL at the transistor level provides practical exposure to the implementation of circuits such as the Phase Frequency Detector, Charge Pump, Loop Filter, Voltage Controlled Oscillator, and Frequency Divider. It also offers valuable insight into important performance parameters including frequency stability, lock time, jitter, phase noise, and loop dynamics.

The motivation for this project is to design and simulate a Phase Locked Loop using UMC 180 nm CMOS technology in the Cadence Virtuoso environment. The project aims to gain practical experience in analog and mixed-signal VLSI design by developing a PLL capable of generating a stable 1.024 GHz output from a 16 MHz reference clock. Through the design, simulation, and analysis of each functional block, this work enhances the understanding of PLL operation and demonstrates its application in high-speed clock generation and frequency synthesis.

## 1.2 Objectives

The primary objective of this project is to design and simulate a **Phase Locked Loop (PLL)** using **UMC 180 nm CMOS technology** in the **Cadence Virtuoso** design environment. The proposed PLL is designed to generate a stable high-frequency output from a low-frequency reference clock while maintaining reliable synchronization and stable loop operation. The project also aims to provide practical knowledge of analog and mixed-signal VLSI design through transistor-level implementation and simulation.

The specific objectives of this project are as follows:

- To study the operating principles and architecture of a Phase Locked Loop (PLL).
- To design the major functional blocks of the PLL, namely the Phase Frequency Detector (PFD), Charge Pump, Passive Loop Filter, Voltage Controlled Oscillator (VCO), and Frequency Divider.
- To implement each functional block using UMC 180 nm CMOS technology in the Cadence Virtuoso environment.
- To integrate all the designed blocks and realize a complete PLL capable of generating an output frequency of approximately 1.024 GHz from a 16 MHz reference clock.

- To analyze the tuning characteristics of the Voltage Controlled Oscillator and determine its gain ( $K_{VCO}$ ).
- To evaluate the performance of the designed PLL in terms of output frequency, lock time, frequency stability, and overall loop operation.
- To verify the functionality of the complete PLL through transistor-level simulations and analyze the obtained results.
- To gain practical experience in analog and mixed-signal CMOS circuit design using the Cadence Virtuoso design platform.

### 1.3 Literature Survey

Phase Locked Loops (PLLs) are widely used in modern communication and digital systems for applications such as frequency synthesis, clock generation, clock recovery, and synchronization. Over the years, several researchers have proposed different PLL architectures to improve power efficiency, reduce phase noise and jitter, achieve wider tuning ranges, and enhance loop stability. A review of the existing literature provides valuable insights into the design methodologies and performance improvements achieved in recent PLL implementations.

#### 1) Power Optimized PLL Implementation in 180 nm CMOS Technology [1]

This work presents a power-efficient PLL implemented using 180 nm CMOS technology for frequency synthesis, clock recovery, and wireless communication applications. The proposed architecture consists of a Phase Frequency Detector (PFD), Charge Pump, Passive Loop Filter, Ring Oscillator-based Voltage Controlled Oscillator (VCO), and Frequency Divider. The PLL achieves a tuning range of 280 MHz to 2.47 GHz with a phase noise of -112.4 dBc/Hz at a 1 MHz offset and consumes only 252  $\mu$ W of power. Although the design offers excellent power efficiency, its operating performance is constrained by the limitations of 180 nm technology. The work provides useful design concepts for developing low-power PLL architectures.

#### 2) A Low Jitter Wide Tuning Range Phase Locked Loop with Low Power Consumption in 180 nm CMOS Technology [2]

This paper proposes a PLL architecture designed to achieve low jitter, wide tuning range, and reduced power consumption. The design employs a D-latch based Phase Frequency Detector, Charge Pump, second-order passive Loop Filter, and a Current Controlled Oscillator (ICO). The PLL provides a tuning range of 420 MHz to 3.1 GHz while maintaining a maximum jitter of 27.1 ps and a power consumption of 343.7  $\mu$ W. The use of a telescopic OTA-based ICO improves tuning range and frequency stability.

However, the VCO remains the dominant source of power consumption, indicating the need for further optimization.

### **3) A CMOS VCO Ring Oscillator Suitable for Wide Supply Range Applications [3]**

This work presents a three-stage CMOS ring oscillator operating over a wide supply voltage range using 110 nm CMOS technology. The oscillator employs an operational amplifier configured as a voltage-controlled current source to maintain a constant bias current despite supply voltage variations. The design successfully operates from 1 V to 3.6 V while maintaining stable oscillation frequency and consistent VCO gain across different process, voltage, and temperature (PVT) conditions. The proposed architecture demonstrates improved supply voltage tolerance and enhanced frequency stability compared with conventional ring oscillators.

### **4) Design and Analysis of Current Starved Ring VCO Targeting SCL 180 nm CMOS Process [4]**

This paper investigates the design of a current-starved ring Voltage Controlled Oscillator intended for low-power PLL applications. The oscillator employs current-limiting transistors to control the propagation delay of each inverter stage, thereby enabling frequency tuning through the control voltage. The design achieves improved power efficiency while maintaining stable oscillation over the desired operating range. Although the architecture is suitable for energy-efficient systems, further improvements are required to reduce phase noise and optimize high-frequency performance.

### **5) Design of a Low Power Wide Range Phase Locked Loop Using 180 nm CMOS Technology [5]**

This paper presents a low-power PLL designed for communication and VLSI applications using 180 nm CMOS technology. The architecture consists of a Phase Frequency Detector, Charge Pump, Passive Loop Filter, Ring Voltage Controlled Oscillator, and Frequency Divider. The proposed PLL provides a wide operating frequency range while maintaining low power consumption and stable loop operation. The design demonstrates that careful optimization of each functional block significantly improves overall PLL performance, making it suitable for frequency synthesis and clock generation applications.

Based on the reviewed literature, it is observed that significant research has been carried out to improve PLL performance in terms of power consumption, tuning range, phase noise, jitter, and locking speed. However, achieving an optimal balance among these parameters remains a challenging task. Motivated by these studies, the present work focuses on the transistor-level design and simulation of a Phase Locked Loop using UMC 180 nm CMOS technology in the Cadence Virtuoso environment. The proposed design emphasizes the implementation, integration, and performance evaluation of all

major PLL building blocks for reliable high-frequency clock generation.

Table 1.1: Comparison of Existing PLL Architectures Reported in Literature

| Ref. | Paper Title                                                                                           | Technology  | Frequency Range    | Power         | Major Contribution                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------|-------------|--------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------|
| [1]  | Power Optimized PLL Implementation in 180 nm CMOS Technology                                          | 180 nm CMOS | 280 MHz – 2.47 GHz | 252 $\mu$ W   | Low-power PLL using a modified ring VCO, passive loop filter, and charge pump for frequency synthesis and wireless applications. |
| [2]  | A Low Jitter Wide Tuning Range Phase Locked Loop with Low Power Consumption in 180 nm CMOS Technology | 180 nm CMOS | 420 MHz – 3.1 GHz  | 343.7 $\mu$ W | Wide tuning range, low jitter, D-latch based PFD, and second-order passive loop filter for improved stability.                   |
| [3]  | A CMOS VCO Ring Oscillator Suitable for Wide Supply Range Applications                                | 110 nm CMOS | 80 MHz             | NR            | Stable VCO operation over a 1 V–3.6 V supply range with improved VCO gain stability and supply noise immunity.                   |
| [4]  | Analysis and Design of Current Starved Ring VCO                                                       | 180 nm CMOS | NR                 | NR            | Current-starved ring VCO providing low-power operation and improved frequency stability for PLL applications.                    |
| [5]  | Design of a Low Power Wide Range Phase Locked Loop Using 180 nm CMOS Technology                       | 180 nm CMOS | 280 MHz – 2.47 GHz | 252 $\mu$ W   | Wide tuning range PLL with low power consumption suitable for communication and frequency synthesis applications.                |

## 1.4 Problem Statement

Modern communication systems, high-speed processors, and digital integrated circuits require accurate and stable clock signals to ensure reliable data processing and synchronization. As the operating frequency of electronic systems continues to increase, conventional clock generation techniques often face challenges such as frequency instability, longer locking time, phase noise, and timing jitter. These issues can significantly affect the performance and reliability of high-speed VLSI systems.

Designing an efficient Phase Locked Loop (PLL) involves achieving a balance between frequency stability, lock time, power consumption, and circuit complexity. The performance of the PLL depends on the proper design and integration of its major functional

blocks, namely the Phase Frequency Detector (PFD), Charge Pump, Passive Loop Filter, Voltage Controlled Oscillator (VCO), and Frequency Divider. Any mismatch or improper operation of these blocks can lead to unstable loop behavior, inaccurate frequency generation, and degraded overall performance.

The problem addressed in this project is to design and simulate a Phase Locked Loop using UMC 180 nm CMOS technology in the Cadence Virtuoso environment. The proposed design aims to generate a stable output frequency of approximately 1.024 GHz from a 16 MHz reference clock using a divide-by-64 feedback architecture. The objective is to achieve reliable frequency synchronization, stable loop operation, and successful frequency multiplication through transistor-level implementation and simulation of all PLL building blocks.

## 1.5 Organization of the Report

This report is organized into five chapters, each focusing on a specific stage of the design, implementation, and evaluation of the proposed Phase Locked Loop (PLL).

Chapter 1 presents the introduction to the project, including the motivation, objectives, literature survey, problem statement, and the overall organization of the report. It provides the background and significance of the proposed work.

Chapter 2 discusses the fundamental concepts of the Phase Locked Loop, its applications, system architecture, mathematical analysis, and design specifications. This chapter establishes the theoretical foundation required for the design of the proposed PLL.

Chapter 3 describes the design and implementation of the individual building blocks of the PLL, namely the Phase Frequency Detector (PFD), Charge Pump, Passive Loop Filter, Voltage Controlled Oscillator (VCO), and Frequency Divider. It also explains the design methodology, circuit implementation, and complete PLL integration.

Chapter 4 presents the simulation results and performance analysis of the designed PLL. The performance of each circuit block and the complete PLL is evaluated using Cadence Virtuoso simulations. The obtained results are discussed in terms of output frequency, lock time, VCO characteristics, and overall loop performance.

Chapter 5 summarizes the work carried out in this project and presents the conclusions drawn from the simulation results. It also discusses the scope for future improvements and possible enhancements to the proposed PLL design.

# Chapter 2

## SYSTEM DESIGN

This chapter presents the theoretical background and system design of the proposed Phase Locked Loop (PLL). It begins with the fundamental concepts of PLL operation, followed by its applications in modern electronic and communication systems. The chapter also discusses the architecture of the proposed PLL, the mathematical principles governing its operation, and the design specifications adopted for this work. Understanding these concepts provides the foundation for the transistor-level implementation of the individual PLL building blocks presented in the subsequent chapter.

sectionIntroduction to Phase Locked Loop

A **Phase Locked Loop (PLL)** is a closed-loop feedback control system that synchronizes the phase and frequency of an output signal with a reference signal. It is one of the most widely used circuits in modern electronic systems due to its ability to generate stable clock signals, perform frequency multiplication, and maintain synchronization between different system components. PLLs are extensively employed in communication systems, microprocessors, wireless transceivers, clock generation circuits, and frequency synthesizers where accurate timing and frequency stability are essential.

A typical PLL consists of five major functional blocks, namely the **Phase Frequency Detector (PFD)**, **Charge Pump (CP)**, **Loop Filter (LF)**, **Voltage Controlled Oscillator (VCO)**, and **Frequency Divider**. The PFD compares the reference clock with the feedback clock and generates control signals based on the phase and frequency difference between them. These control signals are converted into current by the Charge Pump and filtered by the Loop Filter to generate a stable control voltage. The control voltage regulates the oscillation frequency of the VCO, while the Frequency Divider scales the output frequency and feeds it back to the PFD, thereby completing the closed-loop feedback system.

For an Integer-N PLL, the relationship between the reference frequency and output frequency is expressed as

$$f_{out} = N \times f_{ref} \quad (2.1)$$

where,

- $f_{out}$  = Output frequency
- $f_{ref}$  = Reference frequency
- $N$  = Frequency division ratio

For the proposed PLL,

$$f_{ref} = 16 \text{ MHz} \quad (2.2)$$

$$N = 64 \quad (2.3)$$

Hence,

$$f_{out} = 64 \times 16 \text{ MHz} = 1024 \text{ MHz} = 1.024 \text{ GHz} \quad (2.4)$$

The performance of a PLL is primarily evaluated using parameters such as output frequency, locking time, frequency stability, phase noise, jitter, and loop stability. Proper design and integration of each functional block are essential to achieve reliable operation and accurate frequency synthesis. The following sections describe the applications, architecture, mathematical analysis, and design specifications of the proposed Phase Locked Loop.

## 2.1 Applications of Phase Locked Loop

Phase Locked Loops (PLLs) are widely used in modern electronic systems due to their ability to generate stable clock signals, perform frequency synthesis, and maintain phase synchronization. Their capability to automatically track variations in input frequency and phase makes them an essential component in both analog and digital integrated circuits.

One of the major applications of PLLs is in **frequency synthesizers**, where a single low-frequency reference signal is used to generate multiple high-frequency clock signals. This technique is widely employed in wireless communication systems, radio transmitters, and receivers to achieve accurate carrier frequency generation.

PLLs are extensively used in **clock generation and clock distribution** for microprocessors, microcontrollers, and digital signal processors. They multiply the reference clock to the required operating frequency while maintaining synchronization throughout the system, thereby improving overall performance and timing accuracy.

Another important application is in **clock and data recovery (CDR)** circuits used in high-speed communication systems. In these applications, the PLL extracts timing information from incoming serial data streams and regenerates a synchronized clock for reliable data recovery.

PLLs also play a significant role in **wireless communication systems** such as Bluetooth, Wi-Fi, LTE, and 5G networks, where they are employed for carrier generation, modulation, demodulation, and frequency synchronization. Their low phase noise and stable frequency characteristics contribute to improved communication quality.

In addition, PLLs are widely used in **Analog-to-Digital Converters (ADCs)**, **Digital-to-Analog Converters (DACs)**, satellite communication systems, radar systems, instrumentation equipment, and multimedia devices. The ability of a PLL to provide stable and accurate clock signals makes it an indispensable component in modern VLSI applications.

The broad range of PLL applications demonstrates its importance in integrated circuit design. Consequently, designing an efficient PLL with good frequency stability, fast locking characteristics, and reliable operation is essential for meeting the performance requirements of modern electronic systems.

## 2.2 Proposed PLL Architecture

The proposed Phase Locked Loop (PLL) is implemented using a closed-loop feedback architecture to generate a stable high-frequency output from a low-frequency reference clock. The complete PLL is designed using **UMC 180 nm CMOS technology** in the **Cadence Virtuoso** design environment. The architecture consists of five major functional blocks: the **Phase Frequency Detector (PFD)**, **Charge Pump (CP)**, **Passive Loop Filter (LF)**, **Voltage Controlled Oscillator (VCO)**, and **Frequency Divider**. These blocks work together to achieve frequency multiplication and phase synchronization.

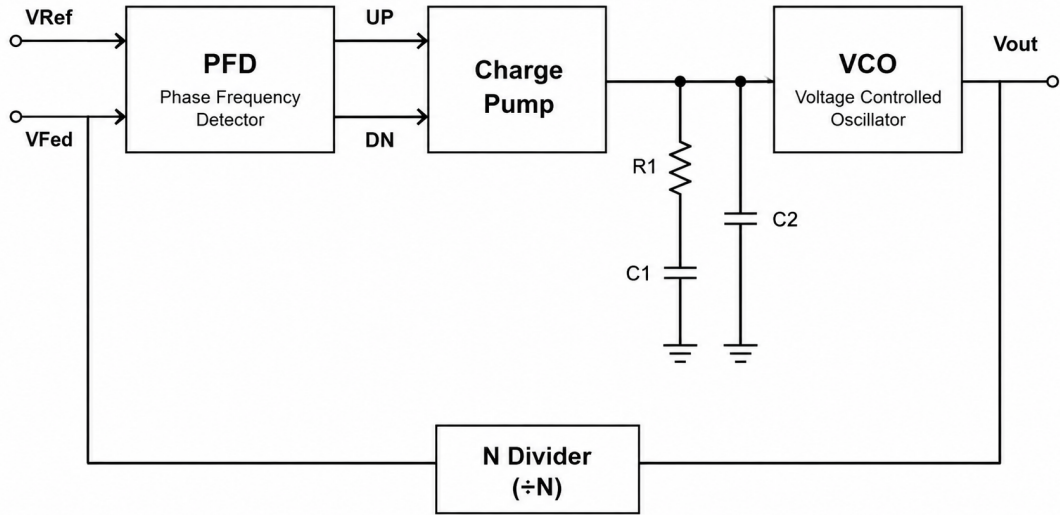


Figure 2.1: Block diagram of the proposed Phase Locked Loop (PLL).

The reference clock of **16 MHz** is applied to the Phase Frequency Detector, where it is continuously compared with the feedback signal obtained from the output of the frequency divider. Based on the phase and frequency difference between these two signals, the PFD generates **UP** and **DOWN** control pulses.

These control signals drive the Charge Pump, which converts the digital pulses into charging and discharging currents. The output current of the Charge Pump is applied to the passive Loop Filter, where it is converted into a smooth control voltage. This control voltage determines the oscillation frequency of the Voltage Controlled Oscillator.

The VCO generates the high-frequency output signal, which is simultaneously provided as the PLL output and applied to the Divide-by-64 Frequency Divider. The divider reduces the output frequency and feeds it back to the PFD, thereby completing the closed-loop feedback path. The loop continuously adjusts the VCO frequency until the feedback signal becomes synchronized with the reference clock.

For the proposed design, the output frequency is determined by

$$f_{out} = N \times f_{ref} \quad (2.5)$$

where  $N$  is the divider ratio.

For

$$N = 64, \quad f_{ref} = 16 \text{ MHz} \quad (2.6)$$

the PLL generates

$$f_{out} = 64 \times 16 \text{ MHz} = 1024 \text{ MHz} = 1.024 \text{ GHz} \quad (2.7)$$

The proposed architecture provides reliable frequency multiplication, stable loop operation, and accurate synchronization, making it suitable for high-speed clock generation and frequency synthesis applications.

## 2.3 Mathematical Analysis of Phase Locked Loop

The operation of a Phase Locked Loop (PLL) can be described mathematically by analysing the relationship between the reference signal, feedback signal, and the Voltage Controlled Oscillator (VCO). The PLL continuously minimizes the phase difference between the reference clock and the feedback clock by adjusting the VCO frequency through a closed-loop feedback mechanism.

The instantaneous phase of a periodic signal is expressed as

$$\phi(t) = \omega t + \phi_0 \quad (2.8)$$

where,

- $\phi(t)$  is the instantaneous phase,
- $\omega$  is the angular frequency (rad/s),
- $t$  is the time (s), and
- $\phi_0$  is the initial phase.

The relationship between angular frequency and frequency is given by

$$\omega = 2\pi f \quad (2.9)$$

where  $f$  represents the signal frequency.

The phase error generated by the Phase Frequency Detector is

$$\phi_e = \phi_{ref} - \phi_{fb} \quad (2.10)$$

where

- $\phi_{ref}$  is the phase of the reference signal,
- $\phi_{fb}$  is the phase of the feedback signal.

The PLL continuously adjusts the VCO frequency until the phase error approaches zero.

For an Integer-N PLL, the output frequency is related to the reference frequency by

$$f_{out} = N \times f_{ref} \quad (2.11)$$

where

- $N$  is the frequency divider ratio,
- $f_{ref}$  is the reference frequency,
- $f_{out}$  is the output frequency.

For the proposed design,

$$N = 64 \quad (2.12)$$

and

$$f_{ref} = 16 \text{ MHz} \quad (2.13)$$

Therefore,

$$f_{out} = 64 \times 16 \text{ MHz} = 1024 \text{ MHz} = 1.024 \text{ GHz} \quad (2.14)$$

The Voltage Controlled Oscillator converts the control voltage generated by the loop filter into the corresponding oscillation frequency. The VCO characteristic can be expressed as

$$f_{VCO} = f_0 + K_{VCO}(V_{CTRL} - V_0) \quad (2.15)$$

where

- $f_0$  is the centre frequency,
- $K_{VCO}$  is the VCO gain,
- $V_{CTRL}$  is the control voltage.

These mathematical relationships form the basis for analysing the behaviour of the proposed PLL and are used in the subsequent chapters for the design and performance evaluation of each functional block.

## 2.4 Design Specifications

The design specifications of the proposed Phase Locked Loop (PLL) were selected based on the desired operating frequency, technology constraints, and performance requirements. These specifications serve as the foundation for designing the individual PLL

building blocks and evaluating the overall system performance. The complete PLL was implemented using UMC 180 nm CMOS technology in the Cadence Virtuoso design environment.

Table 2.1: Design Specifications of the Proposed PLL

| Parameter                         | Value                    |
|-----------------------------------|--------------------------|
| Technology                        | UMC 180 nm CMOS          |
| Supply Voltage ( $V_{DD}$ )       | 1.8 V                    |
| Reference Frequency ( $f_{ref}$ ) | 16 MHz                   |
| Output Frequency ( $f_{out}$ )    | 1.024 GHz                |
| PLL Architecture                  | Integer-N PLL            |
| Divider Ratio ( $N$ )             | 64                       |
| Phase Frequency Detector          | D Flip-Flop Based        |
| Charge Pump Current ( $I_{CP}$ )  | 145 $\mu$ A              |
| Loop Filter Type                  | Passive Second-Order RC  |
| Loop Filter Resistance ( $R$ )    | 45 k $\Omega$            |
| Loop Filter Capacitance ( $C_1$ ) | 6.5 pF                   |
| Loop Filter Capacitance ( $C_2$ ) | 500 fF                   |
| Voltage Controlled Oscillator     | Current-Starved Ring VCO |
| VCO Gain ( $K_{VCO}$ )            | 514 MHz/V                |
| Frequency Divider                 | Divide-by-64             |
| Design Environment                | Cadence Virtuoso         |

The above specifications were used throughout the design and simulation of the proposed PLL. Each functional block was designed to satisfy these requirements before being integrated into the complete closed-loop system. The detailed implementation of the Phase Frequency Detector, Charge Pump, Loop Filter, Voltage Controlled Oscillator, and Frequency Divider is presented in the next chapter.

# Chapter 3

## Design and Implementation of PLL Building Blocks

This chapter presents the transistor-level design and implementation of the proposed Phase Locked Loop (PLL) using UMC 180 nm CMOS technology in the Cadence Virtuoso environment. The complete PLL consists of five major functional blocks: the Phase Frequency Detector (PFD), Charge Pump (CP), Passive Loop Filter (LF), Voltage Controlled Oscillator (VCO), and Frequency Divider. Each block was designed and verified individually before being integrated into the complete PLL architecture.

The design methodology adopted in this work focuses on achieving stable frequency synthesis, reliable phase synchronization, and efficient closed-loop operation. The functionality of each building block is explained in detail, along with the corresponding circuit implementation and design considerations. The complete PLL architecture was obtained by integrating all the individual blocks, enabling the generation of a stable output frequency from the given reference clock.

The following sections discuss the design and implementation of each PLL building block in detail.

### 3.1 Phase Frequency Detector (PFD)

The Phase Frequency Detector (PFD) is the first and one of the most important building blocks of a Phase Locked Loop (PLL). Its primary function is to compare the phase and frequency of the reference clock with those of the feedback clock obtained from the frequency divider. Based on the comparison, the PFD generates two digital output signals, namely **UP** and **DOWN**, which indicate whether the frequency of the Voltage Controlled Oscillator (VCO) should be increased or decreased. These control signals are processed by the Charge Pump and Loop Filter to generate the appropriate control voltage required for frequency correction. A well-designed PFD improves the locking

speed, minimizes phase error, and enhances the overall stability of the PLL.

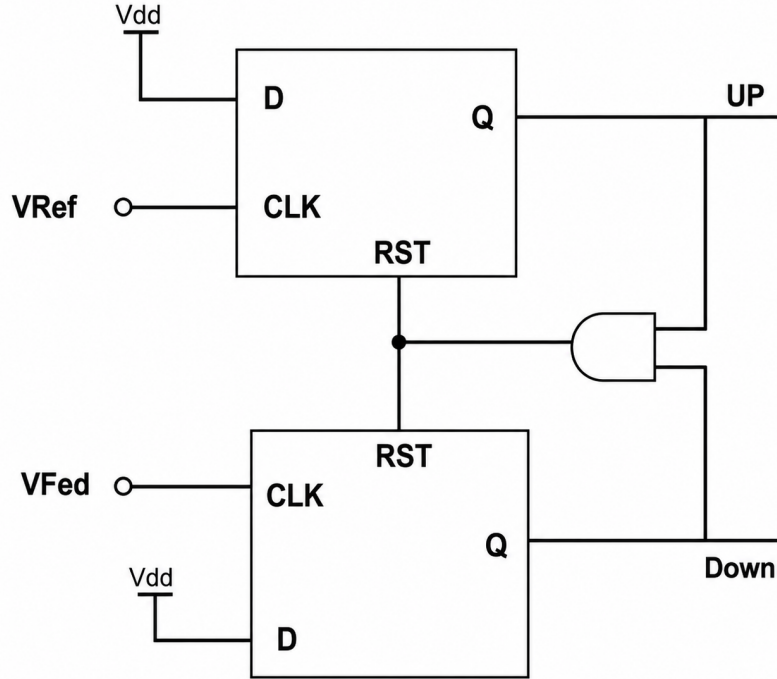


Figure 3.1: Phase Frequency Detector (PFD)

### 3.1.1 D Flip-Flop

The D Flip-Flop is a sequential logic circuit used to store one bit of digital information. It transfers the logic level present at the data input (D) to the output (Q) on the active edge of the clock signal. Due to its edge-triggered operation and reliable data storage capability, the D Flip-Flop is widely used in digital systems such as counters, registers, shift registers, and Phase Locked Loops (PLLs).

In the proposed Phase Frequency Detector (PFD), two positive edge-triggered D Flip-Flops are employed to compare the reference clock and the feedback clock. The data input of each flip-flop is connected to logic HIGH. Whenever a rising edge is detected at the clock input, the corresponding output changes to logic HIGH. The outputs of the two flip-flops generate the **UP** and **DOWN** control signals, which are further processed by the Charge Pump to regulate the frequency of the Voltage Controlled Oscillator (VCO).

The characteristic equation of a D Flip-Flop is given by

$$Q_{n+1} = D \quad (3.1)$$

where,

- $Q_{n+1}$  = Next output state

- $D$  = Data input

The truth table of the D Flip-Flop is shown in Table 3.1

Table 3.1: Truth Table of D Flip-Flop

| Clock          | D | Q (Next State) |
|----------------|---|----------------|
| ↑              | 0 | 0              |
| ↑              | 1 | 1              |
| No Rising Edge | X | Previous State |

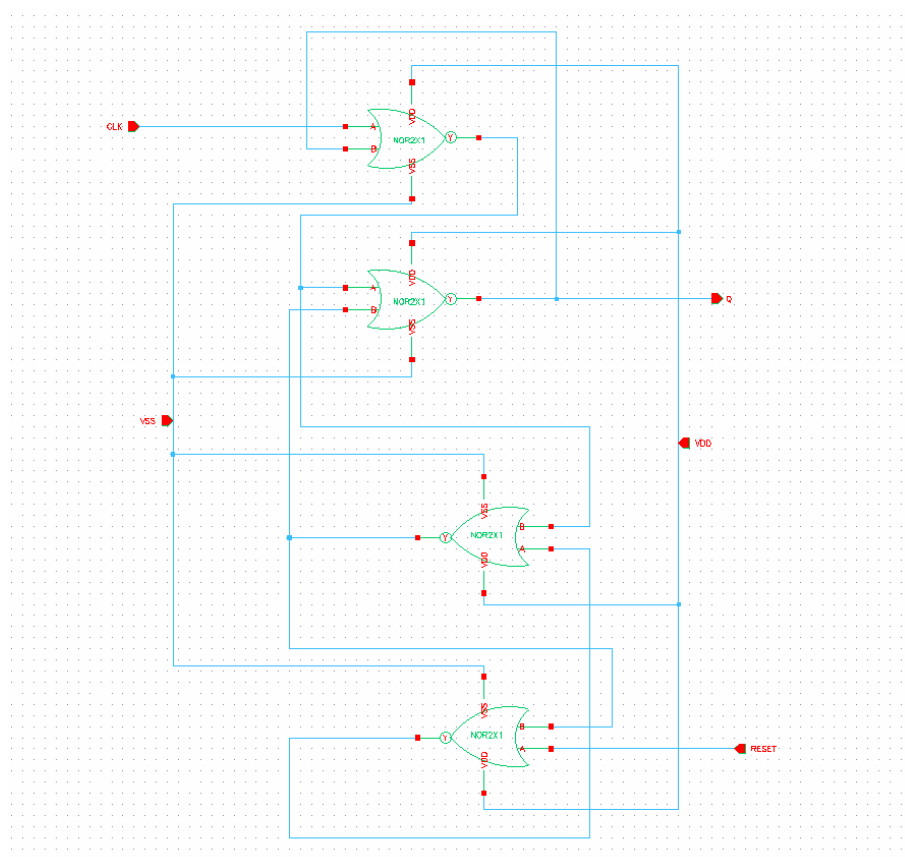


Figure 3.2: Transistor-level schematic of the proposed D Flip-Flop.

The transistor-level D Flip-Flop was designed using CMOS logic in the UMC 180 nm technology library and verified through transient simulations in the Cadence Virtuoso environment. The circuit exhibits reliable edge-triggered operation and forms the basic building block of the proposed Phase Frequency Detector.

### 3.1.2 NOR Gate

The NOR gate is one of the fundamental logic gates used in digital circuit design and is known as a universal gate because any Boolean function can be implemented using only NOR gates. In the proposed Phase Frequency Detector (PFD), NOR gates are used as the basic building blocks for constructing the D Flip-Flops. Their simple CMOS implementation and reliable switching characteristics make them suitable for high-speed digital applications.

A two-input NOR gate produces a logic HIGH output only when both inputs are logic LOW. If either input is HIGH, the output becomes LOW. This logical operation is represented by the Boolean expression

$$Y = \overline{A + B} \quad (3.2)$$

where,

The truth table of the NOR gate is shown in Table 3.2.

Table 3.2: Truth Table of NOR Gate

| Input A | Input B | Output Y |
|---------|---------|----------|
| 0       | 0       | 1        |
| 0       | 1       | 0        |
| 1       | 0       | 0        |
| 1       | 1       | 0        |

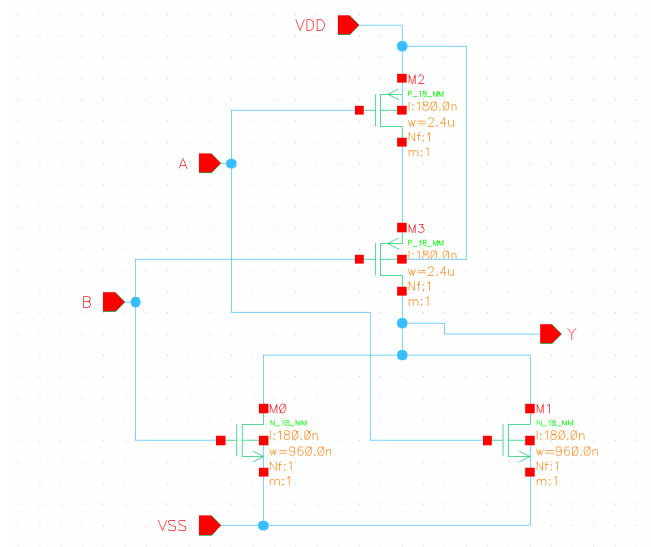


Figure 3.3: Transistor-level schematic of the CMOS NOR gate.

The CMOS implementation of the NOR gate consists of two PMOS transistors connected in series in the pull-up network and two NMOS transistors connected in parallel in the pull-down network. When both inputs are LOW, the PMOS network conducts and pulls the output to logic HIGH. If either input becomes HIGH, the corresponding NMOS transistor conducts and pulls the output to ground, resulting in a logic LOW. The designed NOR gate was implemented using UMC 180 nm CMOS technology and verified through simulation before being incorporated into the D Flip-Flop of the proposed Phase Frequency Detector.

### 3.1.3 AND Gate

The AND gate is a fundamental combinational logic circuit that performs the logical AND operation. In the proposed Phase Frequency Detector (PFD), the AND gate is used to generate the reset signal for the two D Flip-Flops. Its primary function is to detect when both the **UP** and **DOWN** outputs are simultaneously HIGH and immediately reset the flip-flops. This automatic reset mechanism prepares the PFD for the next phase comparison cycle and ensures continuous loop operation.

The Boolean expression of a two-input AND gate is given by

$$Y = A \cdot B \quad (3.3)$$

where,

- $A$  = First input
- $B$  = Second input
- $Y$  = Output

The truth table of the AND gate is shown in Table 3.3.

Table 3.3: Truth Table of AND Gate

| Input A | Input B | Output Y |
|---------|---------|----------|
| 0       | 0       | 0        |
| 0       | 1       | 0        |
| 1       | 0       | 0        |
| 1       | 1       | 1        |

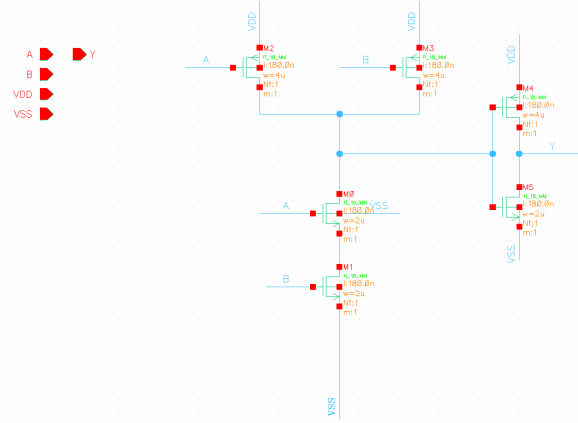


Figure 3.4: Transistor-level schematic of the CMOS AND gate.

The AND gate receives the **UP** and **DOWN** signals generated by the two D Flip-Flops. Under normal operating conditions, only one of these signals is HIGH at a time, keeping the output of the AND gate LOW. When both outputs become HIGH simultaneously, the AND gate generates a reset pulse that clears both flip-flops. This reset action prevents continuous assertion of the UP and DOWN signals, eliminates false triggering, and enables accurate phase and frequency comparison during the next clock cycle.

The CMOS AND gate was implemented using UMC 180 nm CMOS technology in the Cadence Virtuoso environment. The circuit was verified through transient simulations before being integrated into the complete Phase Frequency Detector.

### 3.1.4 Working of the Phase Frequency Detector

The proposed Phase Frequency Detector (PFD) compares both the phase and frequency of the reference clock and the feedback clock obtained from the frequency divider. It generates two output signals, namely **UP** and **DOWN**, which indicate whether the frequency of the Voltage Controlled Oscillator (VCO) should be increased or decreased. The PFD is implemented using two positive edge-triggered D Flip-Flops along with reset logic realized using an AND gate.

The reference clock is applied to the clock input of the first D Flip-Flop, while the feedback clock is connected to the clock input of the second D Flip-Flop. The data inputs of both flip-flops are permanently connected to logic HIGH. Whenever a rising edge appears at either clock input, the corresponding flip-flop output changes to logic HIGH. These outputs form the UP and DOWN control signals.

If the reference clock leads the feedback clock, the UP signal becomes HIGH while the DOWN signal remains LOW. This indicates that the VCO is operating at a lower frequency than required. Consequently, the Charge Pump increases the control voltage, causing the VCO frequency to increase.

Similarly, if the feedback clock leads the reference clock, the DOWN signal becomes HIGH while the UP signal remains LOW. This condition indicates that the VCO frequency is higher than the desired value. The Charge Pump discharges the Loop Filter capacitor, reducing the control voltage and consequently decreasing the VCO frequency.

When both the reference and feedback clocks become synchronized, the UP and DOWN signals are simultaneously reset through the AND gate. The reset pulse clears both D Flip-Flops and prepares the PFD for the next comparison cycle. During the locked condition, only narrow correction pulses are generated to compensate for small phase variations, thereby maintaining synchronization between the reference and feedback signals.

The operating conditions of the proposed Phase Frequency Detector are summarized in Table 3.4.

Table 3.4: Operation of the Phase Frequency Detector

| Reference Clock | Feedback Clock | UP   | DOWN | PLL Action             |
|-----------------|----------------|------|------|------------------------|
| Leads           | Lags           | High | Low  | Increase VCO Frequency |
| Lags            | Leads          | Low  | High | Decrease VCO Frequency |
| Synchronized    | Synchronized   | Low  | Low  | PLL Locked             |

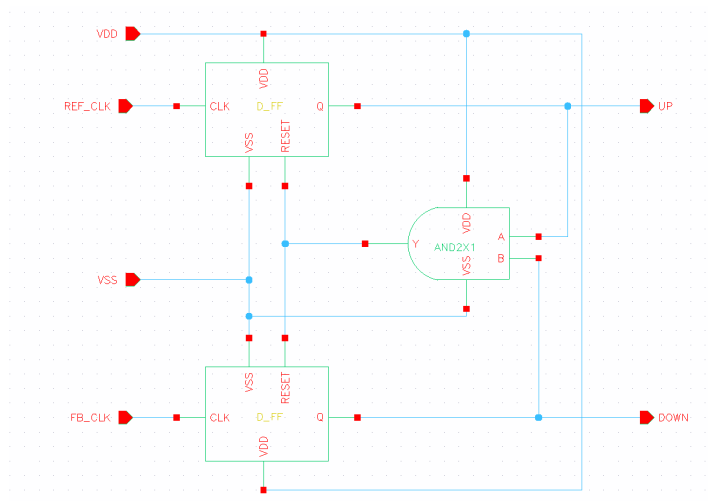


Figure 3.5: Complete transistor-level schematic of the proposed Phase Frequency Detector.

The proposed PFD provides reliable phase and frequency detection with a simple CMOS implementation. Its ability to detect both phase and frequency differences enables faster frequency acquisition, improved loop stability, and accurate synchronization. The transistor-level implementation was verified using Cadence Virtuoso simulations before integration with the Charge Pump in the complete PLL.

## 3.2 Charge Pump (CP)

The Charge Pump (CP) is an essential analog building block of the proposed Phase Locked Loop (PLL). It acts as an interface between the digital Phase Frequency Detector (PFD) and the passive Loop Filter. The Charge Pump converts the digital **UP** and **DOWN** pulses generated by the PFD into proportional charging and discharging currents. These currents are integrated by the Loop Filter to produce a control voltage, which regulates the oscillation frequency of the Voltage Controlled Oscillator (VCO). The performance of the Charge Pump directly influences the locking time, loop stability, phase error, and overall performance of the PLL.

The proposed Charge Pump was designed using CMOS current source and current sink circuits in the UMC 180 nm CMOS technology. Proper matching of the charging and discharging currents minimizes current mismatch, reduces reference spurs, and improves frequency stability. The following subsections describe the design, operation, equations, and implementation of the proposed Charge Pump.

The Charge Pump converts the digital output signals of the Phase Frequency Detector into analog current pulses. Depending on the phase difference between the reference clock and the feedback clock, the PFD generates either an UP pulse or a DOWN pulse. These pulses control the charging and discharging paths of the Charge Pump.

When the UP signal is active, the Charge Pump supplies current to the Loop Filter capacitor, increasing the control voltage applied to the Voltage Controlled Oscillator (VCO). Conversely, when the DOWN signal is active, the Charge Pump removes charge from the Loop Filter capacitor, reducing the control voltage. This continuous charging and discharging action enables the PLL to minimize the phase error and maintain frequency synchronization.

The Charge Pump is designed to deliver a constant current during both charging and discharging operations. Maintaining equal source and sink currents is essential to reduce steady-state phase error and improve loop accuracy. In the proposed design, a Charge Pump current of **145  $\mu\text{A}$**  was selected to provide an appropriate balance between fast locking and stable loop operation.

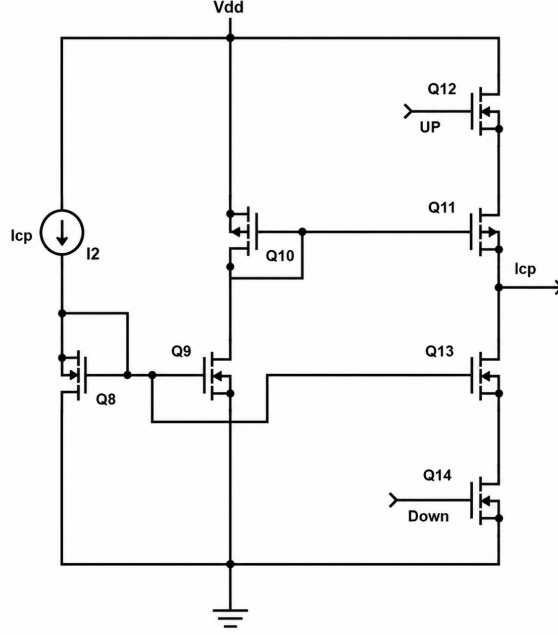


Figure 3.6: Transistor-level schematic of the proposed Charge Pump.

### 3.2.1 Circuit Description

The proposed Charge Pump consists of complementary PMOS and NMOS transistors configured as current source and current sink circuits. The PMOS branch is responsible for supplying current to the Loop Filter during the charging operation, whereas the NMOS branch removes current from the Loop Filter during the discharging operation. The gates of these transistors are controlled by the **UP** and **DOWN** signals generated by the Phase Frequency Detector (PFD).

When the **UP** signal is asserted, the PMOS current source is activated while the NMOS current sink remains OFF. As a result, a constant current flows into the Loop Filter capacitor, increasing the control voltage applied to the Voltage Controlled Oscillator (VCO). Conversely, when the **DOWN** signal is asserted, the PMOS current source is turned OFF and the NMOS current sink is activated. This removes charge from the Loop Filter capacitor, reducing the control voltage and consequently decreasing the oscillation frequency of the VCO.

The Charge Pump is designed such that the charging and discharging currents are closely matched. Proper current matching minimizes current mismatch, reduces reference spurs, and improves the linearity of the PLL. In addition, the use of complementary CMOS transistors provides low static power consumption and high switching efficiency, making the circuit suitable for high-speed and low-power PLL applications.

The transistor dimensions and bias conditions were carefully selected to obtain a Charge Pump current of approximately **145  $\mu\text{A}$** . The designed circuit was implemented

using the UMC 180 nm CMOS technology library in the Cadence Virtuoso environment and verified through transient simulations before integration with the Loop Filter and the remaining PLL building blocks.

### 3.2.2 Circuit Operation

The operation of the proposed Charge Pump is controlled by the **UP** and **DOWN** signals generated by the Phase Frequency Detector (PFD). Based on these control signals, the Charge Pump either charges or discharges the Loop Filter capacitor, thereby adjusting the control voltage applied to the Voltage Controlled Oscillator (VCO). This mechanism enables the PLL to minimize the phase error and achieve frequency synchronization.

#### Charging Mode

When the **UP** signal is HIGH and the **DOWN** signal is LOW, the PMOS current source is activated while the NMOS current sink remains OFF. A constant current flows into the Loop Filter capacitor, increasing the control voltage. As the control voltage increases, the oscillation frequency of the VCO also increases, reducing the phase and frequency difference between the reference and feedback signals.

The charge delivered to the Loop Filter capacitor is given by

$$Q = I_{CP} \times t \quad (3.4)$$

where,

- $Q$  = Charge transferred (C)
- $I_{CP}$  = Charge Pump current (A)
- $t$  = Charging duration (s)

#### Discharging Mode

When the **DOWN** signal is HIGH and the **UP** signal is LOW, the PMOS current source is turned OFF and the NMOS current sink is activated. The stored charge in the Loop Filter capacitor is removed, causing the control voltage to decrease. Consequently, the oscillation frequency of the VCO decreases until the desired frequency is achieved.

The change in the control voltage is expressed as

$$\Delta V = \frac{I_{CP} \times t}{C} \quad (3.5)$$

where,

- $\Delta V$  = Change in control voltage (V)

- $C$  = Loop Filter capacitance (F)

### Locked Condition

When the PLL reaches the locked state, the reference clock and the feedback clock become synchronized. Under this condition, the charging and discharging currents are balanced, resulting in a nearly constant control voltage. The VCO then oscillates at the desired output frequency, and only small correction pulses are generated to compensate for minor phase variations.

The balanced current condition can be expressed as

$$I_{UP} \approx I_{DOWN} \quad (3.6)$$

As a result, the average current flowing into the Loop Filter approaches zero, ensuring stable operation of the PLL with minimal steady-state phase error.

### 3.2.3 Design Equations

The design of the Charge Pump is based on generating equal charging and discharging currents to ensure stable operation of the Phase Locked Loop. The current delivered by the Charge Pump determines the amount of charge transferred to the Loop Filter capacitor, which directly influences the control voltage applied to the Voltage Controlled Oscillator (VCO). Therefore, proper selection of the Charge Pump current is essential for achieving fast lock time, low steady-state phase error, and stable loop dynamics.

The amount of charge transferred during each UP or DOWN pulse is given by

$$Q = I_{CP} \times t \quad (3.7)$$

where,

- $Q$  = Charge transferred (C)
- $I_{CP}$  = Charge Pump current (A)
- $t$  = Pulse duration (s)

The control voltage developed across the Loop Filter capacitor is expressed as

$$V_{CTRL} = \frac{Q}{C} \quad (3.8)$$

Substituting Equation (3.8) into Equation (3.9),

$$V_{CTRL} = \frac{I_{CP} \times t}{C} \quad (3.9)$$

where,

- $V_{CTRL}$  = Control voltage applied to the VCO
- $C$  = Loop Filter capacitance (F)

The average current supplied by the Charge Pump can be expressed as

$$I_{AVG} = I_{UP} - I_{DOWN} \quad (3.10)$$

where

- $I_{UP}$  = Charging current
- $I_{DOWN}$  = Discharging current

For ideal PLL operation,

$$I_{UP} = I_{DOWN} = I_{CP} \quad (3.11)$$

Current matching between the charging and discharging paths minimizes steady-state phase error and improves loop stability. In the proposed design, the Charge Pump current was selected as

$$I_{CP} = 145 \mu A \quad (3.12)$$

This current value provides an appropriate trade-off between fast frequency acquisition and stable closed-loop operation. A larger Charge Pump current decreases the lock time but may increase overshoot, whereas a smaller current improves stability at the expense of slower locking performance.

### 3.2.4 Design Considerations

The performance of the Charge Pump has a significant impact on the overall operation of the Phase Locked Loop. Therefore, several design considerations were taken into account during the implementation to ensure stable loop performance and reliable frequency synchronization.

One of the primary design objectives was to achieve accurate current matching between the PMOS charging branch and the NMOS discharging branch. Any mismatch between these currents introduces a steady-state phase error, increases reference spurs, and degrades the overall performance of the PLL. Therefore, proper transistor sizing and biasing were adopted to maintain nearly identical charging and discharging currents.

Another important consideration was the selection of the Charge Pump current. A current value of **145  $\mu A$**  was chosen to obtain an appropriate balance between locking

speed and loop stability. Increasing the Charge Pump current reduces the lock time by charging the Loop Filter capacitor more rapidly; however, it may also introduce overshoot and instability. Conversely, a smaller current improves stability but increases the time required for the PLL to achieve lock.

The switching characteristics of the Charge Pump were also considered during the design process. Fast switching of the PMOS and NMOS transistors ensures accurate transfer of charge to the Loop Filter and minimizes delay between the Phase Frequency Detector and the Voltage Controlled Oscillator. This contributes to improved dynamic performance and faster frequency acquisition.

Power consumption was another important factor considered in the design. Since the Charge Pump operates continuously as part of the feedback loop, the circuit was implemented using CMOS technology to achieve low static power dissipation while maintaining high switching efficiency. The use of complementary PMOS and NMOS transistors also improves current efficiency and reduces unnecessary power loss.

The proposed Charge Pump was designed and verified using the Cadence Virtuoso environment with the UMC 180 nm CMOS technology library. The circuit demonstrated reliable charging and discharging operation and was successfully integrated with the Passive Loop Filter to generate the control voltage required for stable PLL operation.

### **3.3 Passive Loop Filter (LF)**

The Passive Loop Filter (LF) is an essential component of the proposed Phase Locked Loop (PLL) that connects the Charge Pump to the Voltage Controlled Oscillator (VCO). Its primary function is to convert the charging and discharging current pulses generated by the Charge Pump into a smooth and stable control voltage. This control voltage is applied to the VCO to regulate its oscillation frequency and maintain phase synchronization with the reference signal.

Apart from generating the control voltage, the Loop Filter suppresses high-frequency switching components produced by the Charge Pump and reduces voltage ripple at the VCO input. This improves the stability of the PLL, minimizes phase noise and jitter, and enhances the overall dynamic response of the system. The design of the Loop Filter significantly influences important PLL parameters such as lock time, loop bandwidth, phase margin, and frequency stability.

In the proposed PLL, a second-order passive RC Loop Filter is employed because of its simple structure, low power consumption, and excellent stability. The filter consists of one resistor and two capacitors that together provide the required filtering action while maintaining an appropriate trade-off between locking speed and loop stability. The selected component values are based on the design specifications of the proposed PLL and were optimized to achieve reliable closed-loop operation.

The following subsections describe the design, circuit operation, mathematical analysis, and implementation of the proposed Passive Loop Filter.

### 3.3.1 Introduction

The Passive Loop Filter (LF) is an important analog component of the Phase Locked Loop (PLL) that processes the output current generated by the Charge Pump and converts it into a stable control voltage for the Voltage Controlled Oscillator (VCO). The quality of this control voltage directly influences the frequency stability, locking time, and dynamic response of the PLL. Therefore, the design of the Loop Filter plays a crucial role in determining the overall performance of the system.

In the proposed PLL, a second-order passive RC Loop Filter is employed due to its simple implementation, low power consumption, and good stability characteristics. The filter consists of one resistor and two capacitors arranged to provide both integration and high-frequency noise suppression. The resistor introduces a zero in the loop transfer function, improving the phase margin and loop stability, while the second capacitor introduces a high-frequency pole that attenuates switching noise generated by the Charge Pump.

The Passive Loop Filter smooths the charging and discharging current pulses produced by the Charge Pump, thereby reducing voltage ripple at the VCO input. This results in stable frequency control, reduced timing jitter, and improved phase noise performance. Furthermore, the selected filter parameters determine important PLL characteristics such as loop bandwidth, settling time, and transient response.

For the proposed design, the Loop Filter components were selected as a resistor of **45 k $\Omega$** , a primary capacitor of **6.5 pF**, and a secondary capacitor of **500 fF**. These values were chosen to achieve a suitable compromise between fast locking, stable operation, and reliable frequency synthesis. The transistor-level implementation and performance of the Loop Filter are discussed in the following subsections.

### 3.3.2 Circuit Description

The proposed Passive Loop Filter is a second-order RC network connected between the Charge Pump and the Voltage Controlled Oscillator (VCO). Its primary purpose is to integrate the charging and discharging currents produced by the Charge Pump and generate a smooth control voltage for the VCO. The filter also suppresses high-frequency switching components and minimizes voltage ripple, thereby improving the stability and dynamic performance of the PLL.

The circuit consists of one resistor ( $R$ ) and two capacitors ( $C_1$  and  $C_2$ ). Capacitor  $C_1$  acts as the main integrating capacitor and stores the charge delivered by the Charge Pump. The resistor  $R$  is connected in series with capacitor  $C_2$  and introduces a zero in

the loop transfer function, which improves the phase margin and enhances loop stability. Capacitor  $C_2$  provides an additional high-frequency pole that attenuates switching noise and reduces ripple in the control voltage.

When the Charge Pump supplies current, capacitor  $C_1$  charges gradually, causing the control voltage to increase. Similarly, when the Charge Pump sinks current, the stored charge in  $C_1$  decreases, reducing the control voltage. The combined action of the resistor and capacitors ensures smooth voltage variation while preventing abrupt changes that could lead to oscillations or instability in the PLL.

The values of the Loop Filter components used in the proposed design are listed below:

- Loop Filter Resistance,  $R = 45\text{ k}\Omega$
- Primary Capacitor,  $C_1 = 6.5\text{ pF}$
- Secondary Capacitor,  $C_2 = 500\text{ fF}$

These component values were selected to provide an appropriate balance between loop bandwidth, phase margin, settling time, and frequency stability. The designed Loop Filter effectively converts the pulsed output of the Charge Pump into a stable control voltage required for reliable operation of the Voltage Controlled Oscillator.

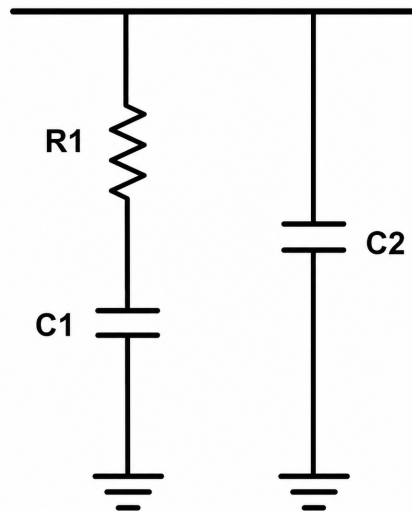


Figure 3.7: Loop Filter

### 3.3.3 Circuit Operation

The operation of the Passive Loop Filter is based on integrating the charging and discharging currents supplied by the Charge Pump to produce a smooth control voltage

for the Voltage Controlled Oscillator (VCO). The Loop Filter eliminates high-frequency switching components from the Charge Pump output and ensures that only the low-frequency control signal is applied to the VCO. This enables the PLL to achieve stable operation and accurate frequency synchronization.

### Charging Operation

When the **UP** signal generated by the Phase Frequency Detector is HIGH, the Charge Pump supplies a constant current to the Loop Filter. The current charges the main capacitor ( $C_1$ ), causing the control voltage to increase gradually. As the control voltage rises, the oscillation frequency of the VCO also increases until the phase error between the reference and feedback signals is minimized.

The voltage across the capacitor during charging is given by

$$V_{CTRL} = \frac{1}{C_1} \int I_{CP} dt \quad (3.13)$$

where

- $V_{CTRL}$  = Control voltage applied to the VCO
- $I_{CP}$  = Charge Pump current
- $C_1$  = Main Loop Filter capacitor

### Discharging Operation

When the **DOWN** signal becomes HIGH, the Charge Pump removes current from the Loop Filter capacitor. As the stored charge decreases, the control voltage is reduced. This causes the oscillation frequency of the VCO to decrease until the required output frequency is obtained.

The decrease in capacitor voltage is expressed as

$$\Delta V = \frac{I_{CP} \times t}{C_1} \quad (3.14)$$

where  $t$  represents the discharge duration.

### Steady-State Operation

When the PLL reaches the locked condition, the charging and discharging currents become nearly equal. Consequently, the average current flowing into the Loop Filter approaches zero, and the control voltage remains almost constant. The resistor ( $R$ ) and capacitor ( $C_2$ ) suppress high-frequency ripple and stabilize the control voltage, ensuring reliable operation of the VCO with minimum phase fluctuations.

The average current at the locked condition is given by

$$I_{AVG} = I_{UP} - I_{DOWN} \approx 0 \quad (3.15)$$

As a result, the VCO operates at the desired output frequency while maintaining phase synchronization with the reference clock. The second-order passive Loop Filter therefore provides stable control voltage generation, reduced ripple, improved phase margin, and enhanced loop stability, making it suitable for high-speed PLL applications.

### 3.3.4 Design Calculations

The design of the passive Loop Filter was carried out using the standard second-order Charge Pump PLL design methodology presented in the NPTEL course on Analog Integrated Circuit Design. The loop filter parameters were selected to satisfy the required loop bandwidth, phase margin, and frequency stability while ensuring fast locking and stable closed-loop operation.

The design specifications of the proposed PLL are summarized as follows:

$$f_{ref} = 16 \text{ MHz} \quad (3.16)$$

$$f_{out} = 1.024 \text{ GHz} \quad (3.17)$$

$$N = \frac{f_{out}}{f_{ref}} \quad (3.18)$$

Substituting the design values,

$$N = \frac{1024}{16} = 64 \quad (3.19)$$

Thus, the divider ratio of the proposed PLL is

$$N = 64 \quad (3.20)$$

The gain of the Voltage Controlled Oscillator obtained from simulation is

$$K_{VCO} = 514 \text{ MHz/V} \quad (3.21)$$

The Phase Frequency Detector and Charge Pump together provide the phase detector gain

$$K_{PD} = 23.08 \text{ } \mu\text{V/rad} \quad (3.22)$$

The Charge Pump current selected for the proposed design is

$$I_{CP} = 145 \mu A \quad (3.23)$$

Based on the above design parameters, the passive second-order Loop Filter components were determined following the standard NPTEL design procedure. The loop filter resistor is calculated using

$$R = \frac{2\zeta\omega_n N}{K_{PD}K_{VCO}} \quad (3.24)$$

where

- $\zeta$  = Damping factor
- $\omega_n$  = Natural frequency of the PLL
- $N$  = Divider ratio
- $K_{PD}$  = Phase detector gain
- $K_{VCO}$  = VCO gain

Similarly, the main Loop Filter capacitor is obtained from

$$C_1 = \frac{K_{PD}K_{VCO}}{\omega_n^2 N} \quad (3.25)$$

The ripple suppression capacitor is selected according to the NPTEL design guideline as

$$C_2 \approx \frac{C_1}{10} \quad (3.26)$$

Using the above equations and the required loop specifications, the final component values selected for the proposed PLL are

$$R = 45 \text{ } k\Omega \quad (3.27)$$

$$C_1 = 6.5 \text{ } pF \quad (3.28)$$

$$C_2 = 500 \text{ } fF \quad (3.29)$$

The obtained values satisfy the required loop bandwidth and provide adequate phase margin for stable PLL operation. The resistor introduces a compensating zero that improves transient response, while the second capacitor introduces a high-frequency pole that suppresses switching ripple generated by the Charge Pump. The selected Loop Filter therefore provides a smooth control voltage to the Voltage Controlled Oscillator, resulting in reliable frequency locking and stable closed-loop performance.

### 3.3.5 Design Considerations

The design of the Passive Loop Filter plays a crucial role in determining the dynamic performance and stability of the Phase Locked Loop (PLL). Appropriate selection of the filter components is necessary to achieve the desired loop bandwidth, phase margin, locking time, and frequency stability. During the design process, several factors were considered to ensure reliable operation of the proposed PLL.

The resistor and capacitor values were selected based on the design methodology presented in the NPTEL PLL design guidelines. The resistor introduces a zero in the loop transfer function, which improves the phase margin and enhances the transient response of the PLL. The primary capacitor acts as the integrating element of the filter and generates the control voltage required by the Voltage Controlled Oscillator (VCO). The secondary capacitor introduces a high-frequency pole that suppresses switching ripple produced by the Charge Pump and improves the stability of the control voltage.

A larger Loop Filter capacitance improves noise suppression and reduces voltage ripple at the VCO input. However, increasing the capacitance also increases the lock time of the PLL. On the other hand, a smaller capacitance provides faster frequency acquisition but may result in higher control voltage ripple and reduced stability. Therefore, an appropriate compromise between locking speed and frequency stability was achieved by selecting suitable capacitor values.

Similarly, the resistance value directly influences the loop bandwidth and damping factor of the PLL. A larger resistance increases the phase margin and improves stability but reduces the loop bandwidth. Conversely, a smaller resistance increases the loop bandwidth and decreases the lock time, although excessive reduction may lead to oscillations and unstable loop behavior.

For the proposed PLL, the selected component values are

$$R = 45 \text{ k}\Omega \quad (3.30)$$

$$C_1 = 6.5 \text{ pF} \quad (3.31)$$

$$C_2 = 500 \text{ fF} \quad (3.32)$$

These values provide an appropriate balance between loop bandwidth, phase margin, transient response, and frequency stability. The designed Loop Filter was implemented using passive components in the Cadence Virtuoso environment and verified through simulation before integration with the Voltage Controlled Oscillator. The simulation results confirmed that the filter produces a smooth control voltage with minimal ripple, enabling reliable frequency locking and stable operation of the complete PLL.

## 3.4 Voltage Controlled Oscillator (VCO)

The Voltage Controlled Oscillator (VCO) is the core building block of the proposed Phase Locked Loop (PLL). It converts the control voltage generated by the Loop Filter into an oscillating output signal whose frequency varies according to the applied control voltage. The VCO determines the output frequency of the PLL and therefore has a significant influence on important performance parameters such as tuning range, phase noise, jitter, frequency stability, and locking speed.

In the proposed PLL, a current-starved ring oscillator is employed due to its compact structure, low power consumption, wide tuning range, and ease of CMOS implementation. The oscillation frequency is controlled by regulating the current flowing through each delay stage. As the control voltage changes, the propagation delay of the inverter stages changes, thereby varying the oscillation frequency.

The designed VCO operates from a 1.8 V supply and generates an output frequency of approximately 1.024 GHz. The VCO gain obtained from simulation is 514 MHz/V. The following subsections describe the design, operation, mathematical analysis, and implementation of the proposed Voltage Controlled Oscillator.

### 3.4.1 Introduction

A Voltage Controlled Oscillator (VCO) is an electronic oscillator whose output frequency is controlled by an input voltage. In a Phase Locked Loop, the control voltage is generated by the Loop Filter and continuously adjusted until the PLL achieves phase and frequency synchronization. The VCO therefore acts as the frequency generating element of the PLL.

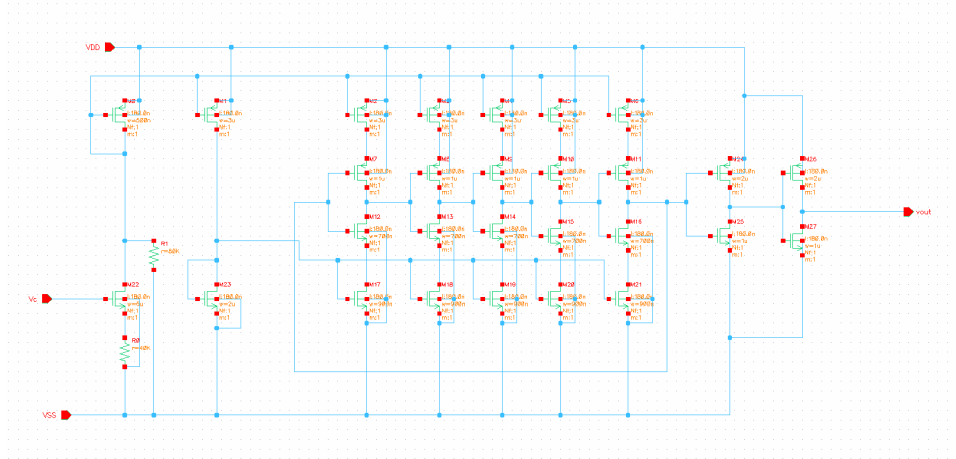
Among various VCO architectures, the current-starved ring oscillator is widely used because of its simple implementation, low power consumption, and wide tuning capability. Unlike LC oscillators, ring oscillators occupy smaller silicon area and are well suited for CMOS integrated circuit implementation.

The proposed VCO consists of multiple current-starved inverter stages connected in a closed loop. The oscillation frequency is controlled by varying the charging and discharging current through each inverter stage using the control voltage generated by the Loop Filter.

### 3.4.2 Current-Starved Ring Oscillator

The proposed VCO is implemented using a current-starved ring oscillator. Each inverter stage contains PMOS and NMOS current-limiting transistors that regulate the current flowing through the inverter. The amount of current supplied to the inverter depends on the control voltage.

When the control voltage increases, more current flows through the inverter stages, reducing the propagation delay and increasing the oscillation frequency. Similarly, when the control voltage decreases, the available current reduces, increasing the propagation delay and lowering the oscillation frequency.



### 3.4.3 Circuit Description

The control voltage generated by the Loop Filter is applied to the gates of the current-limiting transistors. This voltage controls the current flowing through each inverter stage and consequently determines the propagation delay of the ring oscillator.

### 3.4.4 Circuit Operation

When the control voltage increases, the PMOS and NMOS current-limiting transistors conduct more current, reducing the propagation delay of each stage. Consequently, the oscillation frequency increases.

When the control voltage decreases, the available current reduces, increasing the propagation delay and decreasing the oscillation frequency.

The oscillation continues because the output of the final inverter stage is fed back to the input of the first stage. The odd number of inverter stages introduces a phase shift of  $180^\circ$ , while the propagation delay provides an additional  $180^\circ$  phase shift, satisfying the Barkhausen oscillation criterion.

### 3.4.5 Design Equations

The oscillation frequency of a ring oscillator is given by

$$f_{osc} = \frac{1}{2Nt_d} \quad (3.33)$$

where

- $N$  = Number of inverter stages
- $t_d$  = Propagation delay of each stage

The propagation delay is approximately

$$t_d = \frac{CV}{I} \quad (3.34)$$

where

- $C$  = Load capacitance
- $V$  = Supply voltage
- $I$  = Bias current

Substituting,

$$f_{osc} = \frac{I}{2NCV} \quad (3.35)$$

The VCO gain is defined as

$$K_{VCO} = \frac{\Delta f}{\Delta V_{CTRL}} \quad (3.36)$$

### 3.4.6 Design Calculations

The VCO was designed to operate from a 1.8 V supply and generate an output frequency close to 1.024 GHz.

The gain of the proposed VCO is calculated from the frequency versus control voltage characteristics.

$$K_{VCO} = \frac{\Delta f}{\Delta V} \quad (3.37)$$

From the simulation results,

$$K_{VCO} = 514 \text{ MHz/V} \quad (3.38)$$

The obtained tuning characteristic confirms that the proposed VCO provides adequate frequency sensitivity for the PLL while maintaining stable oscillation throughout the desired operating range.

### 3.4.7 Design Considerations

Several factors were considered during the design of the proposed VCO. The number of delay stages was selected to achieve stable oscillation while maintaining low power consumption. Current-starving transistors were properly sized to provide a wide tuning range and good linearity.

The transistor dimensions were optimized to minimize phase noise and improve frequency stability. The bias current was carefully selected to achieve the desired oscillation frequency without excessive power dissipation.

The proposed current-starved ring oscillator offers a compact layout, wide tuning range, low power consumption, and compatibility with CMOS technology, making it suitable for high-speed PLL applications.

## 3.5 Frequency Divider

The Frequency Divider is an essential digital building block of the proposed Phase Locked Loop (PLL). Its primary function is to divide the high-frequency output of the Voltage Controlled Oscillator (VCO) by a fixed division ratio and feed the reduced frequency back to the Phase Frequency Detector (PFD). The feedback mechanism enables the PLL to continuously compare the divided output with the reference clock and maintain phase and frequency synchronization.

In the proposed PLL, a Divide-by-64 Frequency Divider is implemented using cascaded True Single Phase Clock (TSPC) D Flip-Flops. The TSPC architecture is selected because of its high-speed operation, low power consumption, reduced transistor count, and compatibility with CMOS technology. The divider operates reliably at gigahertz frequencies while maintaining stable timing characteristics.

The following subsections describe the design, implementation, mathematical analysis, and operation of the proposed Frequency Divider.

### 3.5.1 Introduction

The Frequency Divider forms the feedback path of the Phase Locked Loop by reducing the output frequency of the Voltage Controlled Oscillator before it is applied to the Phase Frequency Detector. The divider ratio determines the multiplication factor of the PLL and therefore directly affects the generated output frequency.

A digital frequency divider is commonly implemented using cascaded flip-flops, where each stage divides the input frequency by two. By cascading multiple divide-by-two stages, higher division ratios can be easily achieved. In the proposed design, six TSPC D Flip-Flops are connected in cascade to realize a Divide-by-64 frequency divider.

The TSPC architecture provides high operating speed and low dynamic power consumption, making it suitable for high-frequency PLL applications implemented in CMOS technology.

### 3.5.2 Circuit Description

The proposed Divide-by-64 Frequency Divider consists of six cascaded TSPC D Flip-Flops. Each flip-flop divides the frequency of its input signal by two. The output of one stage is connected to the clock input of the subsequent stage, resulting in successive frequency division.

The first stage receives the oscillation signal generated by the Voltage Controlled Oscillator. After passing through six divide-by-two stages, the output frequency becomes sixty-four times lower than the VCO frequency. This divided signal is then fed back to the Phase Frequency Detector for phase comparison with the reference clock.

The Frequency Divider was implemented using UMC 180 nm CMOS technology and verified through transient simulations in the Cadence Virtuoso environment.

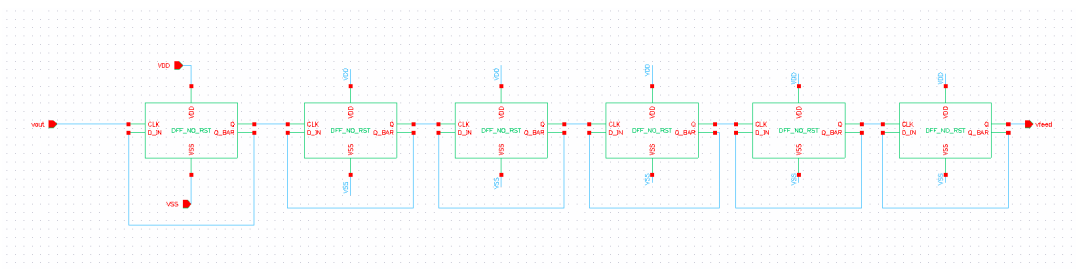


Figure 3.9: Transistor-level schematic of the proposed Divide-by-64 Frequency Divider.

### 3.5.3 True Single Phase Clock (TSPC) D Flip-Flop

The True Single Phase Clock (TSPC) D Flip-Flop is the fundamental building block of the proposed Frequency Divider. Unlike conventional master-slave flip-flops, the TSPC architecture requires only a single clock signal for operation, thereby reducing clock loading, power consumption, and propagation delay.

The TSPC D Flip-Flop operates using dynamic CMOS logic and is capable of functioning at very high frequencies. These characteristics make it highly suitable for high-speed frequency divider applications in modern Phase Locked Loops.

The transistor-level implementation of the TSPC D Flip-Flop was designed using UMC 180 nm CMOS technology and validated through transient simulations before being incorporated into the complete Divide-by-64 Frequency Divider.

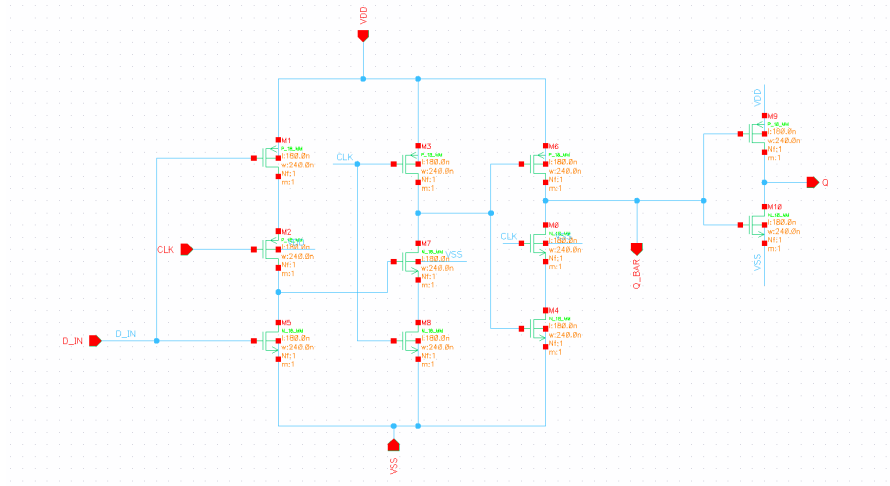


Figure 3.10: Transistor-level schematic of the TSPC D Flip-Flop.

### 3.5.4 Circuit Operation

The operation of the Frequency Divider is based on successive frequency division using cascaded TSPC D Flip-Flops. Each flip-flop divides the input clock frequency by two, producing an output frequency equal to one-half of its input.

The overall division ratio is given by

$$N = 2^m \quad (3.39)$$

where

- $N$  = Division ratio
- $m$  = Number of cascaded divide-by-two stages

For the proposed design,

$$m = 6 \quad (3.40)$$

Therefore,

$$N = 2^6 = 64 \quad (3.41)$$

The output frequency of the divider is expressed as

$$f_{div} = \frac{f_{VCO}}{64} \quad (3.42)$$

Substituting the designed VCO frequency,

$$f_{div} = \frac{1.024 \text{ GHz}}{64} = 16 \text{ MHz} \quad (3.43)$$

The divided output frequency is equal to the reference clock frequency, enabling the Phase Frequency Detector to compare both signals and maintain stable PLL operation.

### 3.5.5 Design Considerations

The Frequency Divider was designed to operate reliably at gigahertz frequencies while maintaining low power consumption and high switching speed. The TSPC architecture was selected because it requires only a single clock phase, resulting in reduced clock loading and simplified circuit implementation.

The transistor dimensions were carefully optimized to minimize propagation delay and ensure reliable operation under high-frequency conditions. Particular attention was given to reducing dynamic power consumption while maintaining sufficient noise margins for stable digital operation.

The complete Divide-by-64 Frequency Divider was verified through transient simulations in the Cadence Virtuoso environment. The simulation results confirmed correct frequency division and stable operation before integration into the complete PLL feedback loop.

## 3.6 Complete PLL Integration

The complete Phase Locked Loop (PLL) is realized by integrating the Phase Frequency Detector (PFD), Charge Pump (CP), Passive Loop Filter (LF), Voltage Controlled Oscillator (VCO), and Divide-by-64 Frequency Divider into a closed-loop feedback system. Each building block was individually designed and verified using the Cadence Virtuoso environment before being interconnected to obtain the complete PLL architecture.

The reference clock of 16 MHz is applied to the Phase Frequency Detector, where

it is continuously compared with the feedback signal obtained from the output of the Frequency Divider. Based on the phase and frequency difference between these two signals, the PFD generates the UP and DOWN control signals.

The UP and DOWN signals drive the Charge Pump, which converts the digital pulses into proportional charging and discharging currents. These currents are applied to the Passive Loop Filter, where they are integrated to produce a smooth control voltage. The generated control voltage regulates the oscillation frequency of the Voltage Controlled Oscillator.

The Voltage Controlled Oscillator generates the high-frequency output signal of approximately 1.024 GHz. This output is simultaneously provided as the PLL output and applied to the Divide-by-64 Frequency Divider. The divider reduces the VCO frequency to 16 MHz and feeds it back to the Phase Frequency Detector, thereby completing the feedback loop.

The closed-loop operation continuously adjusts the VCO frequency until the divided output frequency becomes equal to the reference frequency. Once synchronization is achieved, the PLL enters the locked state and maintains a stable output frequency with minimal phase error.

The complete PLL was implemented using UMC 180 nm CMOS technology in the Cadence Virtuoso design environment. The integrated design was verified through transient and frequency-domain simulations to evaluate its locking behavior, frequency stability, and overall system performance.

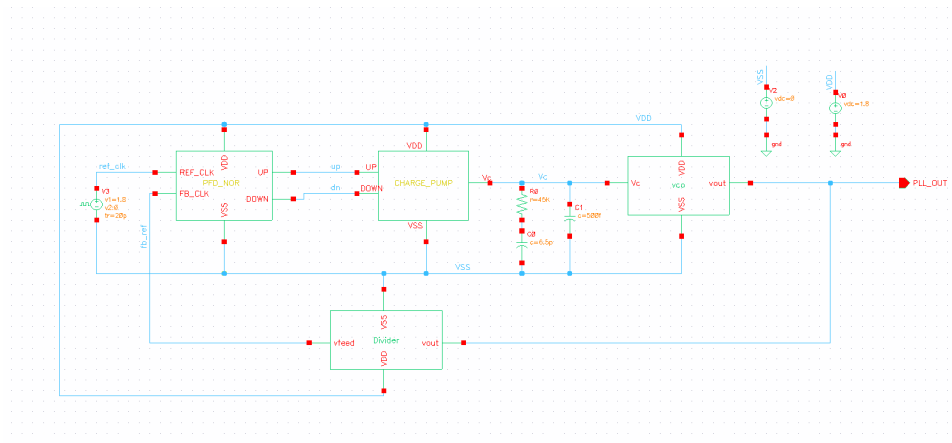


Figure 3.11: Complete transistor-level schematic of the proposed Phase Locked Loop.

The successful integration of all the designed building blocks demonstrates the functionality of the proposed PLL architecture. The complete PLL was simulated under the specified operating conditions to verify frequency multiplication, loop stability, and locking characteristics. The simulation results and performance analysis of each circuit block and the complete PLL are presented in the next chapter.

# Chapter 4

## Simulation Results and Discussion

This chapter presents the simulation results of the proposed Phase Locked Loop (PLL) implemented using UMC 180 nm CMOS technology in the Cadence Virtuoso environment. Each functional block was verified individually before integrating the complete PLL. The simulation results are presented and discussed to validate the functionality and performance of the proposed design.

### 4.1 Phase Frequency Detector Results

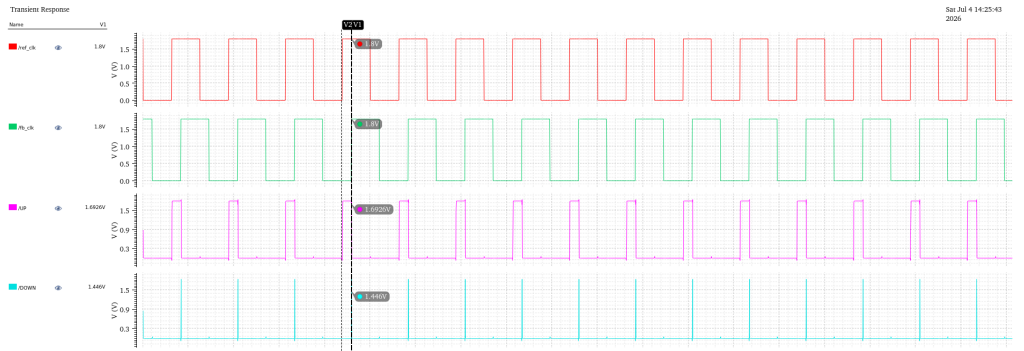


Figure 4.1: Simulation result of the Phase Frequency Detector.

The simulated waveform confirms the correct operation of the Phase Frequency Detector. The circuit successfully generates the UP and DOWN signals according to the phase and frequency difference between the reference and feedback clocks. The obtained results verify accurate phase detection and proper functionality of the proposed PFD.

## 4.2 PFD with Charge Pump Results

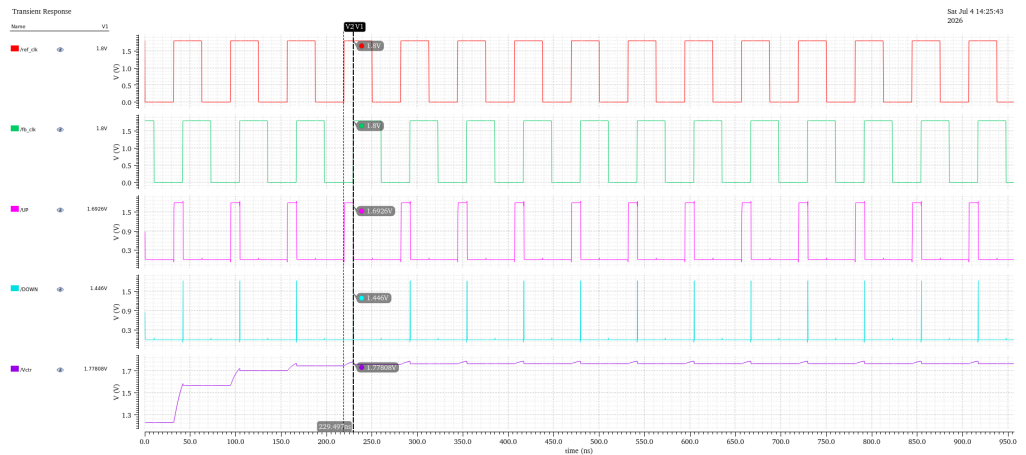


Figure 4.2: Simulation result of the Phase Frequency Detector with Charge Pump.

The simulation demonstrates the interaction between the Phase Frequency Detector and the Charge Pump. The generated UP and DOWN pulses are converted into charging and discharging currents, producing the control voltage required for the Loop Filter. The results confirm proper operation of the Charge Pump and its successful integration with the PFD.

### 4.3 Voltage Controlled Oscillator Results

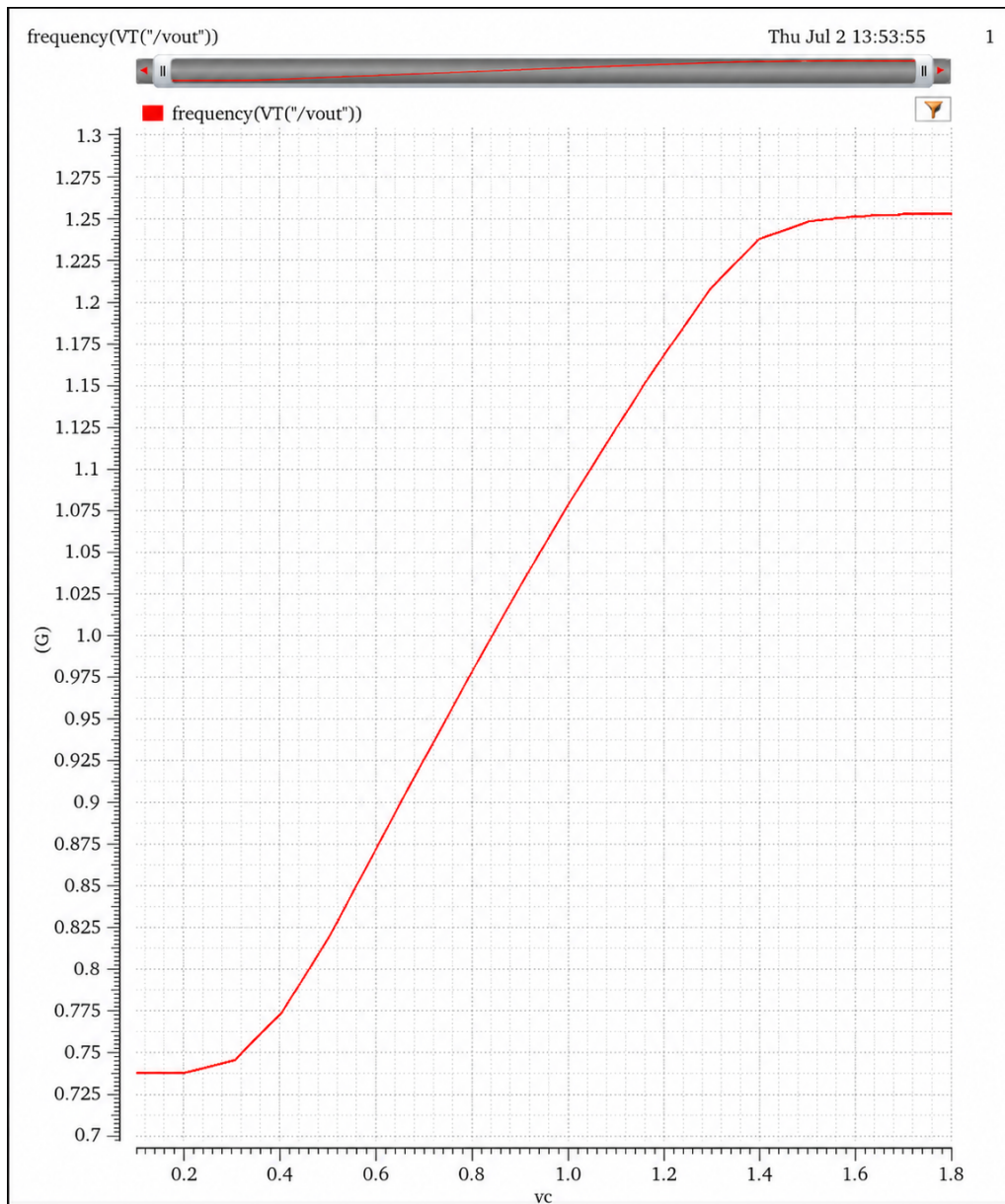


Figure 4.3: Frequency versus control voltage characteristics of the VCO.

The VCO simulation verifies that the oscillation frequency varies with the applied control voltage. The obtained frequency tuning characteristic demonstrates stable operation throughout the required operating range and confirms the effectiveness of the proposed current-starved ring oscillator.

## 4.4 VCO Gain ( $K_{VCO}$ )

Table 4.1: Calculation of VCO Gain

|    | vc       | deriv(fre.../vout''')) |
|----|----------|------------------------|
| 1  | 200.0E-3 | 6.528E6                |
| 2  | 300.0E-3 | 77.57E6                |
| 3  | 400.0E-3 | 308.8E6                |
| 4  | 500.0E-3 | 471.7E6                |
| 5  | 600.0E-3 | 531.1E6                |
| 6  | 700.0E-3 | 537.7E6                |
| 7  | 800.0E-3 | 530.7E6                |
| 8  | 900.0E-3 | 513.8E6                |
| 9  | 1.000    | 490.7E6                |
| 10 | 1.100    | 468.7E6                |
| 11 | 1.200    | 444.2E6                |
| 12 | 1.300    | 410.2E6                |
| 13 | 1.400    | 284.5E6                |
| 14 | 1.500    | 99.76E6                |
| 15 | 1.600    | 31.53E6                |
| 16 | 1.700    | 12.29E6                |
| 17 | 1.800    | 6.114E6                |

The VCO gain was determined from the simulated frequency tuning characteristics. The calculated gain is approximately **514 MHz/V**, which satisfies the design requirement and provides sufficient frequency sensitivity for stable PLL operation.

## 4.5 Frequency Divider Results



Figure 4.4: Simulation result of the Divide-by-64 Frequency Divider.

The Frequency Divider correctly divides the VCO output frequency by a factor of 64. The simulation verifies stable digital operation and confirms that the divided output matches the reference frequency required for the feedback path of the PLL.

## 4.6 Complete PLL Results

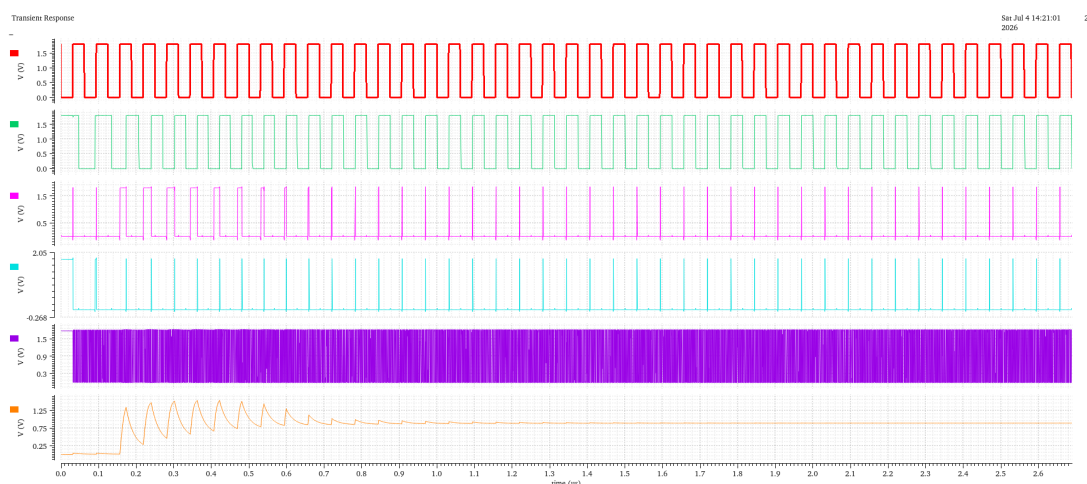


Figure 4.5: Simulation result of the complete Phase Locked Loop.

The complete PLL simulation confirms successful integration of all functional blocks. The output waveform demonstrates stable locking and reliable frequency multiplication from the 16 MHz reference clock to an output frequency of approximately 1.024 GHz.

## 4.7 Frequency Response Analysis

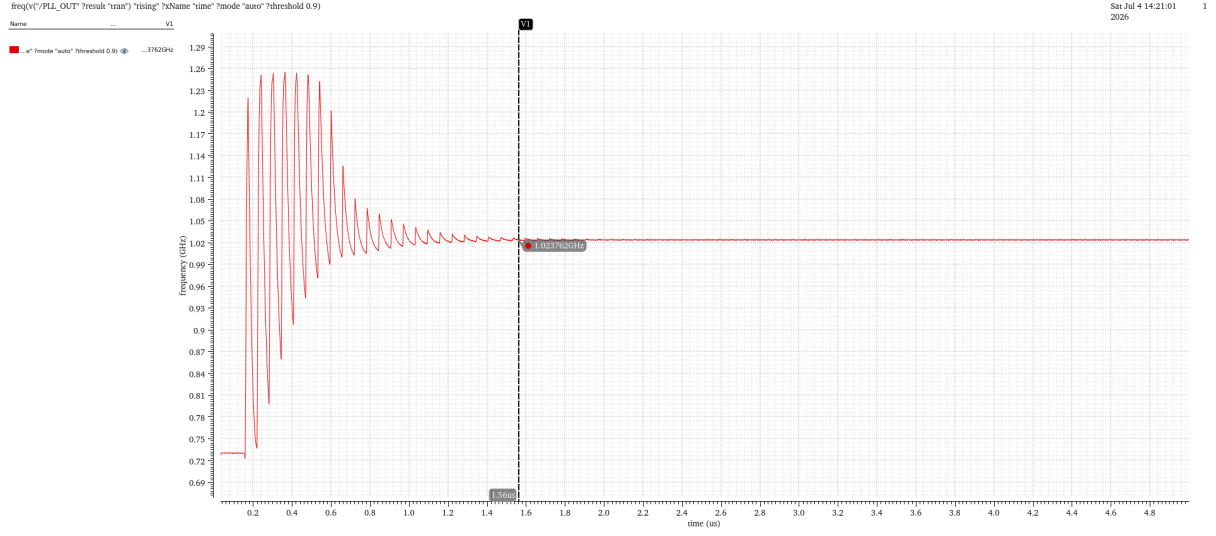


Figure 4.6: Frequency response of the proposed PLL.

The frequency response analysis verifies the stability of the proposed PLL. The obtained loop gain and phase margin indicate stable closed-loop operation and satisfactory dynamic performance under the specified operating conditions.

## 4.8 Performance Summary

Table 4.2 summarizes the important performance parameters of the proposed Phase Locked Loop (PLL). The simulation results indicate that the designed PLL successfully meets the required specifications for frequency multiplication and stable operation. The obtained output frequency is close to the target frequency of 1.024 GHz, while the measured loop gain, phase margin, lock time, phase noise, and RMS jitter demonstrate satisfactory closed-loop performance. These results validate the effectiveness of the proposed PLL architecture implemented using UMC 180 nm CMOS technology.

Table 4.2: Performance Summary of the Proposed PLL

| Parameter                         | Value           |
|-----------------------------------|-----------------|
| Technology                        | UMC 180 nm CMOS |
| Supply Voltage ( $V_{DD}$ )       | 1.8 V           |
| Reference Frequency ( $f_{ref}$ ) | 16 MHz          |
| Output Frequency ( $f_{out}$ )    | 1.024 GHz       |
| Divider Ratio ( $N$ )             | 64              |
| Charge Pump Current ( $I_{CP}$ )  | 145 $\mu$ A     |
| VCO Gain ( $K_{VCO}$ )            | 514 MHz/V       |
| Loop Bandwidth                    | 3.56 MHz        |
| Phase Margin                      | 21.086°         |
| Loop Gain                         | 44.19 dB        |
| Lock Time                         | 1.562 $\mu$ s   |
| Phase Noise @ 1 MHz Offset        | −88.68 dBc/Hz   |
| RMS Jitter                        | 75 ps           |

The simulation results confirm that all the individual building blocks operate correctly and that the integrated PLL achieves stable frequency locking. The obtained performance demonstrates that the proposed design is suitable for high-speed clock generation and frequency synthesis applications. Furthermore, the successful implementation of the PLL using UMC 180 nm CMOS technology validates the effectiveness of the adopted design methodology and confirms the reliability of the proposed architecture.

# Chapter 5

## Conclusion and Future Scope

### 5.1 Conclusion

This work presented the design and simulation of a Phase Locked Loop (PLL) using UMC 180 nm CMOS technology in the Cadence Virtuoso environment. The proposed PLL consists of a Phase Frequency Detector (PFD), Charge Pump (CP), Passive Loop Filter (LF), Current-Starved Ring Voltage Controlled Oscillator (VCO), and a Divide-by-64 Frequency Divider. Each building block was designed, simulated, and verified individually before integrating them into the complete PLL architecture.

The simulation results demonstrate that the proposed PLL successfully performs frequency multiplication from a 16 MHz reference clock to an output frequency of approximately 1.024 GHz. The VCO exhibits a gain of 514 MHz/V, while the integrated PLL achieves stable locking with satisfactory loop gain, phase margin, lock time, phase noise, and RMS jitter. The Frequency Divider correctly provides the required feedback signal, enabling reliable phase and frequency synchronization throughout the closed-loop operation.

The obtained results validate the adopted design methodology and confirm that the proposed PLL architecture provides stable operation, reliable frequency synthesis, and efficient performance. The implementation using UMC 180 nm CMOS technology demonstrates that the designed PLL is suitable for clock generation, frequency synthesis, and communication system applications.

### 5.2 Future Scope

Although the proposed PLL achieves the desired design objectives, several improvements can be considered in future work to further enhance its performance. Advanced CMOS technologies such as 90 nm, 65 nm, or 45 nm can be adopted to reduce power consumption, improve operating speed, and decrease chip area. The passive Loop Filter may be

replaced with an active Loop Filter to provide greater design flexibility and improved loop dynamics.

The phase noise and timing jitter of the VCO can be further minimized by optimizing the oscillator architecture or by employing differential ring oscillators or LC-based oscillators. Adaptive biasing techniques and digital calibration methods can also be incorporated to improve frequency stability under process, voltage, and temperature (PVT) variations.

In addition, the proposed Integer-N PLL can be extended to a Fractional-N PLL to achieve finer frequency resolution and greater flexibility for modern wireless communication systems. These improvements would make the PLL suitable for high-performance applications such as RF transceivers, high-speed serial communication, clock and data recovery circuits, and System-on-Chip (SoC) designs.

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